



DEPARTMENT OF ELECTRICAL AND POWER ENGINEERING

ET 8303: ELECTROMAGNETIC THEORY II

Individual Open Test I [15 Marks]

Start on 12/11/2024 at 16:00 Hrs.

End on 2/11/2024 Submission deadline at 16:00 Hrs.

NOTE: (YOUR WORK SHOULD BE HAND WRITTEN)

PART- A

1. State Faraday's law of induction.
2. State lenz's law
3. Give the equation of transformer emf
4. What is motional electric field?
5. What is motional emf?
6. What is the emf produced by moving loop in time varying field?
7. What is time harmonic field?
8. Write Maxwell's equation in point and integral form for good conductors
9. What Is significance of displacement current density?
10. What are the different ways of EMF generation? Explain with the governing equations and suitable practical examples.
11. Derive General field relation for time varying electric and magnetic fields using Maxwell's' equations.

PART- B

1. Define a Wave.
2. Mention the properties of uniform plane wave.
3. Write down the wave equation for $\vec{E}$ and $\vec{H}$ in free space.

4. Write down the wave equation for $\vec{E}$ and $\vec{H}$ in a conducting medium
5. Define intrinsic impedance or characteristic impedance.
6. Calculate the characteristic impedance of free space.
7. Define propagation constant.
8. Define skin depth.
9. Define Pointing vector.
10. State Poynting Theorem.
11. What is lossy dielectric medium?
12. Define Polarization.
13. Define Circular Polarization.
14. Define Elliptical polarization.
15. Define Linear Polarization.

PART- C

1. (a). Assume that $\vec{E}$ and $\vec{H}$ waves, traveling in free space, are normally Incident on the interface with a perfect dielectric with $\epsilon_r = 3$. calculate the magnitudes of incident, reflected and transmitted $\vec{E}$ and $\vec{H}$ waves at the interface.
 (b). A plane wave propagating through a medium with $\epsilon_r = 8$, $\mu_r = 2$ has $\vec{E} = 0.5 \sin (10^8 t - z) \vec{a}_z$ V/m. Determine
 - i. The loss tangent
 - ii. Wave impedance
 - iii. Wave velocity
2. The electric field in free space is given by $\vec{E} = 50 \cos (10^8 t + \beta x) \vec{a}_y$ V/m
 - i) Find the direction of wave propagation
 - ii) Calculate β and the time it takes to travel a distance of $\lambda/2$
 - iii) Sketch the wave at $t=0$, $T/4$ and $T/2$.